Model Provider Documentation
Supporting Document `IndigoCarbon_US-1_2023_0049`

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Contents

1 Definitions **1**
1.1 Technical Terms 1
1.2 Projects 1

2 Hardware and Software Setup Overview **1**

3 Work-Flow Step-By-Step **2**

4 Summary of Changes **5**

1 Definitions

Brief definition of technical terms and projects below.

1.1 Technical Terms

- GCP – Google Cloud Platform
- API – Application Protocol Interface
- GUI – Graphical User Interface

1.2 Projects

- API GUI Project – Web application (DayCent-CR API Version 2.0.0). This interface is used so the user can create an account and interact with the API endpoints if they don't plan on using their own API.
- API Controller Project – Backend web application (DayCent-CR API Version 2.0.0). This program's job is to help process the run queue.
- Model Translator – (DayCent-CR API Version 2.0.0) Contains DLLs that are used to convert the user input files to DayCent-CR input files. Also responsible for validating errors related to user inputs.
- DayCent-CR Service – Backend application (DayCent-CR API Version 2.0.0). Listens for model run request, runs the model and stores results.
  - DayCent-CR model is run by this application.

2 Hardware and Software Setup Overview

All hardware used to run servers are hosted in the cloud on the GCP. The API and DayCent-CR Model use the following GCP applications for hosting.

- GCP Application Engine
  - Host API GUI application (Dynamically scales as needed)
  - Host API Controller application (Dynamically scales as needed)
- Postgres DB (Automatically backed up)
  - Used for API user data
  - Lookup values
  - Soil components data
- Google Cloud Storage Bucket (automatically backed up, protected and encrypted)
  - Stores user run files
  - Stores user results
  - Weather data for DayCent-CR
- Compute Engine (dynamically scales as needed)
  - Runs on-demand DayCent-CR Service
    - DayCent-CR Model runs inside this service

3 Work-Flow Step-By-Step

- User account creation
  - If an account does not already exist, the user must create an account at api.soilmetrics.eco to gain access to the API.
  - Once the account has been created, API technical staff grant user access to the API and subsequent models.
- Batch runs upload
  - Run files can be uploaded in 2 ways.
  - By using the "upload file" feature provided by the API GUI.
  - Or directly via API using post request and standard authentication tokens.
  - The user has the option to upload files in either be XML or JSON format. Files can be uploaded as any combination of individual files or zip folders.
  - During upload process, if any stage 1 errors are detected, the API will return an error message to the user with more information. If there are no errors, the files will be compressed and uploaded to the active queue located in the Google Cloud Storage Bucket.
  - The user's run file id and queue position are recorded in the User Database. Note, the User Database only stores information about the user and their runs. Neither the run records or results themselves are stored in this database.
- Run queue processing
  - The API controller continually processes available runs in the run queue according to their queue position.
- Parsing User Runs
  - Once the API Controller pulls in a run from the queue, the API Controller uses the Model Translator to parse the user input files, translating them to DayCent-CR schedule files. This is done in parallel; each file is parsed on their own thread.
- Data Input – Data used in user input file

The model translator creates the following data parameters that are used from the user input file. The source of inputs for these data parameters are described in `IndigoCarbon_US-1_2023_`0030. Preparing to use the DayCent-CR Version 1.1.0 model is described in `IndigoCarbon_US-1_2023_0073`. Preparation and submission of inputs to the API are described in `IndigoCarbon_US-1_2023_0070`.

- Geometric Location
- MLRA parcel is in
- Elevation of parcel
- Weather Data of parcel
- Soils Data
- Soil Date
- Name of Soil
- Weather soil is hydric or not
- Initial soil SOC Stock at 30cm
- Soil Layer Information from 0cm – 210cm
  - Bulk Density
  - Coarse Fragments
  - Sand
  - Clay
  - Organic Fraction
  - PH
- Historic crop management – Crop management before 1980
- Historic crop management – Was this parcel ever enrolled in CRP or not
- Historic crop management – Crop management from 1980 – 2000
  - Tillage from 1980 - 2000
- Crop management current period – Crop management for time period from 2000 to before scenario period
- Crop management scenario period – After current period
- Crop management data used for current period and scenario period
  - Planting event year
  - Harvest(s) event dates
  - Harvest event type
  - Harvest event yield amount
  - Harvest event straw stover hay removal percentage
  - Tillage event date
  - Tillage event type
  - Nitrogen application event date
  - Nitrogen application event amount
  - Whether or not this nitrogen application was applied with EEP
  - Manure event application

  - Manure event application date
  - Manure event application amount
  - Manure event percent N
  - Manure event C to N ratio
  - Burn event date
  - Liming event date
  - Liming event method
  - Liming event Rate
  - Grazing event start date
  - Grazing event end date
  - Grazing rest period
  - Grazing utilization percentage
  - Irrigation event start date
  - Irrigation event end date
  - Irrigation event auto or not
  - Irrigation event soil threshold percent
  - Irrigation event inches
  - Irrigation event days between irrigation
  - Is this crop a second crop
  - Is this crop a third crop
  - Is this crop a cover crop
- Data Input – Schedule file creation

The DayCent-CR schedule files are created in order by date from user input file data.

- Schedule file dependencies
  - Site file
    - Created from user soil data
  - .100 files
    - Static set of parameterized files used to run DayCent-CR
  - Weather file
    - From user for DayCent-CR
  - Equilibrium Spin up
    - Lookup schedule file based on parcel location and historic management data
  - Base History
    - Lookup schedule file based on parcel location and historic management data
  - Modern History
    - Lookup schedule file based on parcel location and historic management data
  - Current Period

        * DayCent-CR events created from current management data
    - Scenario Period
        * DayCent-CR events created from scenario management data
- Data Input Error Validation Errors – Model Translator
    - During the data input translation process, the Model Translator checks for errors at each user input. Any caught validation errors are collected and returned to the API Controller.
    - Validation errors returned to the API controller are saved as error logs in the User Database. Error logs are available to user immediately via the API.
    - Validation errors are also returned to the user in an error document after processing cycle is complete.
- Data Inputs Sent to DayCent-CR Service
    - If there are no data validation errors returned by the Model Translator, the API Controller sends the generated DayCent-CR inputs to the DayCent-CR Service.
    - One DayCent-CR Service is created for each input file dynamically, as needed.
    - Once the generated DayCent-CR inputs are received by the DayCent-CR Service from the API Controller, the DayCent-CR Service begins running each schedule file in order from earliest time period.
    - If the Monte Carlo option is selected by the user, the scenario period is re-run 176 times with adjustments to parameters per the monte carlo run instructions, as summarized in `IndigoCarbon_US-1_2023_0065`, and also described in `IndigoCarbon_US-1_2023_0046`.
    - DayCent-CR outputs are saved directly to GCP Cloud storage from DayCent-CR Service.
- DayCent-CR Model Errors
    - During model execution, if errors are detected during the model run, error logs are created and sent back to the API Controller.
    - If the API controller receives errors from the DayCent-CR Service, those errors are stored as error logs in the user database.
        * Errors logs are immediately available for the user to lookup.
        * Errors are also complied in an error document and sent to the user once processing cycle is complete.
    - The DayCent-CR Service attempts to detect any thrown errors or abnormal executions by the model.
- Returning Outputs
    - Once all model runs have completed for a file, the DayCent-CR Service notifies the API Controller.
    - The API controller updates the status of the run file in the User Database to finished.
    - The API Controller continually monitors all finished files and attempts to deliver the results to the user.
    - Results are delivered in 2 potential ways.
        * Email address. User specifies email address when uploading run file for processing.
        * URL webhook. User specifies URL webhook when uploading run file for processing.

4 Summary of Changes

Section	Date	Summary of Changes
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